CHAPTER 09

Carboxylic Acid Derivatives

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

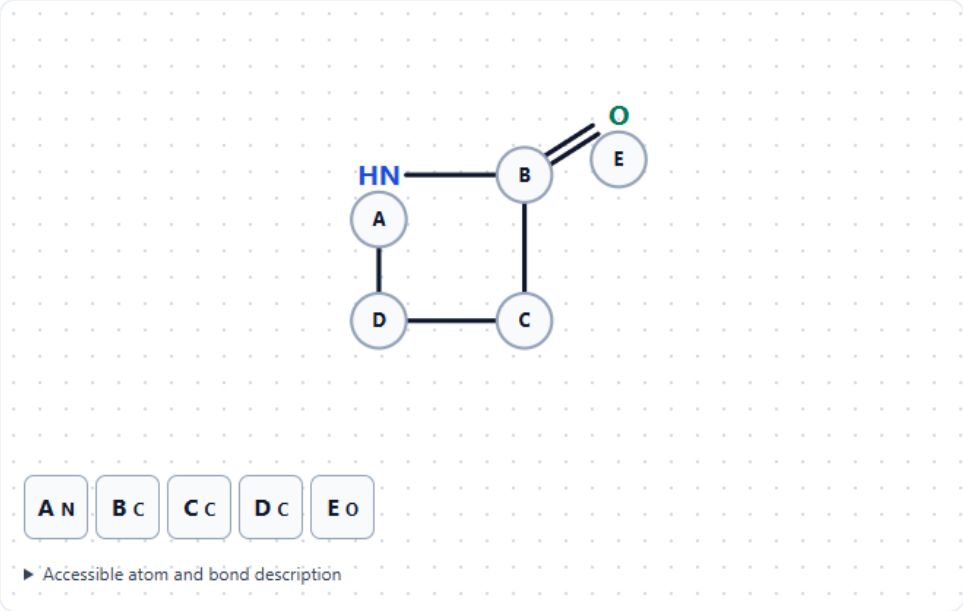
Source snapshot SHA-256: 6155664634c695f24c00b63203875a747c2b8d959f43daca2ab73998d6e79301

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

cyclic-derivative | Independent exercise

Screenshot: cyclic-derivative-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=100.00, y=60.00; charge=0; explicit H override=none

1 (B): C; x=185.00, y=60.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=145.00; charge=0; explicit H override=none

3 (D): C; x=100.00, y=145.00; charge=0; explicit H override=none

4 (E): O; x=240.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 0, 1], [1, 4, 2]]

cyclic-derivative:stage-1 | Select the ring nitrogen and carbonyl carbon.

Answer type: atoms

cyclic-derivative:stage-2 | How many atoms are in the ring?

Answer type: number | unit: ring atoms

cyclic-derivative:stage-3 | The functional-group family is…

Answer type: choice

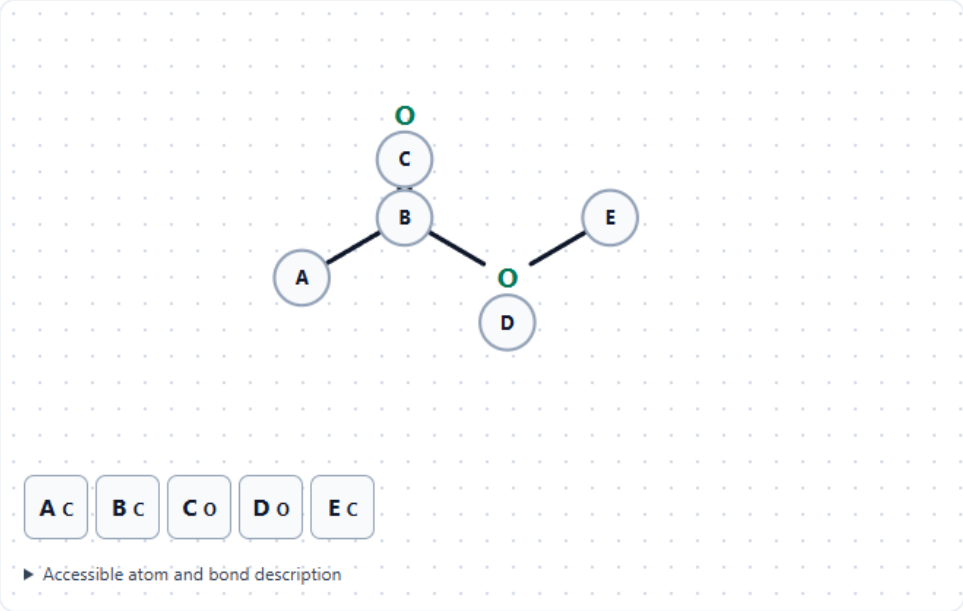
Option 0 (A): Lactam

Option 1 (B): Lactone

Option 2 (C): Ether

acyl-order | Independent exercise

Screenshot: acyl-order-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1]]

acyl-order:stage-1 | Select the electrophilic acyl carbon.

Answer type: atoms

acyl-order:stage-2 | After addition, what restores the carbonyl?

Answer type: choice

Option 0 (A): Collapse of a tetrahedral intermediate with leaving-group departure

Option 1 (B): Free rotation around C=O

Option 2 (C): Loss of the methyl carbon as methane

acyl-order:stage-3 | Among comparable ordinary anhydrides, esters, and amides, which is least reactive toward acyl substitution?

Answer type: choice

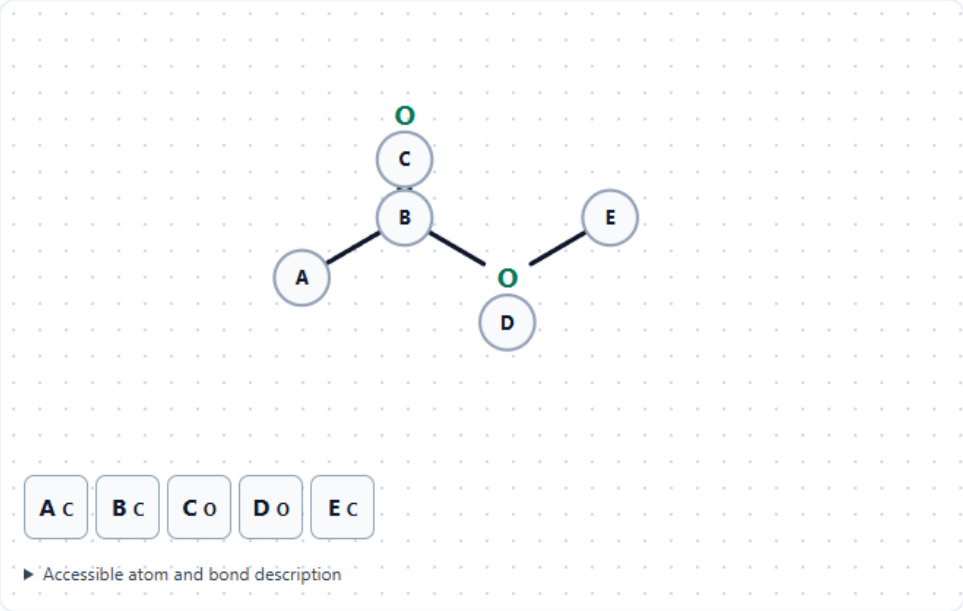
Option 0 (A): Anhydride

Option 1 (B): Ester

Option 2 (C): Amide

hydrolysis-products | Independent exercise

Screenshot: hydrolysis-products-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1]]

hydrolysis-products:stage-1 | Select the O-bound methyl carbon that leaves with the alkoxy fragment.

Answer type: atoms

hydrolysis-products:stage-2 | Under excess aqueous NaOH, the major acid-derived product is…

Answer type: choice

Option 0 (A): Acetate (ethanoate)

Option 1 (B): Neutral ethanoic acid

Option 2 (C): Ethanal

hydrolysis-products:stage-3 | Identify the alcohol produced from the O-methyl group (name or condensed formula).

Answer type: text